09 · Problem 4: Why the Known Patches Keep the Split

People have noticed problems 1–3 before. Each popular fix improves something, but each **keeps two separate "brains"**: one to talk, one to point at memory.

Step 1: What happens

Look at each patch and ask two questions: *Does it still need a separate encoder at search time?* and *Does it add work?*

Patch	What it does	Separate encoder still needed?	Extra work
Query rewriting	LLM writes a self-contained query using the dialogue history.	Yes	None beyond generation
HyDE (Gao et al., 2022)	LLM writes a *fake answer document*, and that is embedded instead of the query. Fake answers look more like real documents.	Yes	Much more text to generate
Hybrid search	Combine keyword search (e.g. BM25) with vector search.	Yes, plus a keyword index	A second index and a way to merge results
Bigger embedders	Use a stronger encoder.	Yes	More compute per query

Patch	What it does	Separate encoder still needed?	Extra work
LLM2Vec / NV-Embed	Turn an LLM *into* an embedding model (bidirectional attention, special pooling, contrastive training).	Yes. It is a second model, or a second forward pass in "embedding mode".	A full extra pass

Step 2: Why it hurts

Every patch treats the symptom (bad or thin query text) and leaves the structure (throw away the LLM's understanding, re-read the text) in place. Some make Problems 1 and 2 **worse**: HyDE generates more text, hybrid adds a second index.

Step 3: A concrete example

With HyDE, the query "cat food" might become a 100-token fake paragraph: *"Choosing a food for your cat depends on age and health. Senior cats with kidney issues benefit from…"*. The fake paragraph is then embedded. Quality may improve, but you now pay for 100 generated tokens **and** the embedder pass.

Step 4: The closest relatives

LLM2Vec and NV-Embed deserve special attention, because they also use an LLM's hidden states for embeddings. The difference is **when**:

	LLM2Vec / NV-Embed	This paper
Whose hidden states?	A modified LLM running specially in embedding mode	The *same* agent LLM, during its normal generation
Extra forward pass?	Yes	No, the states are reused
Extra network	Pooling or a full model	~25M-parameter head
Goal	A better embedding model	No embedding model at search time

Step 5: What a real fix would need

Stop re-reading. Use the representation that already exists at the moment the agent decides to search, and add as little new machinery as possible.

→ That is the core insight in 11.